

Flora Survey Methodologies

A how-to guide to the field methods — when to use each, exactly how to run it, and worked examples.

DOCUMENT CONTROL & REVIEW			
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1.0	—	Initial topic note.	
2.0	Aug 2026	Rebuilt as a how-to guide with per-method step-by-steps and worked examples; added legislative context and perspectives.	

HOW TO READ THIS DOCUMENT

- Process / how-to
- Legislative / assessment
- Definitions / concepts
- Pitfalls / caution
- Worked example

LEARNING OBJECTIVES

- Describe the main survey methods and when each is used
- Run each method correctly, step by step
- Judge and demonstrate adequacy of coverage, replication and timing
- Select and combine methods for a real brief

LEGISLATIVE & ASSESSMENT CONTEXT

- BAM 2020** prescribes the plot type (20×20 m + 50 m transect) and the minimum number of plots per vegetation zone by area.
- NSW & Commonwealth threatened-species survey guidelines** set the expected method, effort and timing for particular species — your targeted searches must meet them.
- EPBC survey guidelines** apply where a nationally listed species is involved.
- “Adequate survey effort”** must be demonstrable — the method you choose is what the consent authority reviews first.

How it links: the BAM tells you *that* you must sample each zone with plots; the survey guidelines tell you *how much* effort a target species needs; together they define whether your method is “adequate”. Choose the method to satisfy both, or the report can be sent back with an RFI.

DEFINITIONS — KEY TERMS

Random meander — an unstructured, wandering search that maximises the species list and rare-plant detection.

Targeted search — a meander concentrated in the habitat of a specific threatened species, in its season.

Quadrat / plot — a fixed sample area (BAM: 20×20 m) for quantitative, repeatable composition and cover.

Transect — a line (BAM: 50 m) along which structure or change is sampled.

Survey adequacy — whether coverage, replication and timing were sufficient to reliably detect what is present.

METHOD 1 — RANDOM MEANDER (HOW TO)

Use it for: building the full species inventory and general detection across a site or zone.

- 1 Divide the site into the vegetation zones/habitats to be covered; allocate meander time to each by area and complexity.
- 2 Walk an irregular path that samples every microhabitat — rocky rises, drainage lines, tracks, canopy gaps — not just the easy ground.
- 3 Record every species new to the list; hand-lens and photograph anything uncertain; log GPS for threatened finds.
- 4 Track and record time and route (GPS tracklog) so effort can be shown; continue until new species taper off.

Worked example. A 6 ha box-gum woodland lot: you split it into 3 zones, spend ~45 min/ha meandering each, deliberately looping through a rocky rise and a drainage line. You log a 2.1 km tracklog and 92 species over 4.5 hours — the tracklog and species accumulation are your evidence the meander was adequate.

METHOD 2 — TARGETED THREATENED-SPECIES SEARCH (HOW TO)

Use it for: demonstrating presence/absence of specific listed species from your target list.

- 1 For each target, read its habitat and flowering window; map the suitable habitat on site from the desktop review.
- 2 Time the visit to the flowering window; search the mapped habitat intensively (parallel sweeps at close spacing).
- 3 Record presence with count, phenology, GPS and photos; record the search effort even where the species is absent.
- 4 Check the effort against the species' survey guideline; add repeat visits if the guideline requires them.

Worked example. Target: *Swainsona recta* (Small Purple-pea), flowers Oct–Nov. You map 1.4 ha of suitable open box-gum woodland, return on 22 October, and walk 5 m-spaced sweeps. You find 37 plants across two patches, record counts, phenology (full flower), GPS polygons and photos — a defensible presence record with mapped effort.

METHOD 3 — QUADRAT / BAM PLOT (HOW TO)

Use it for: quantitative composition and condition data for the BAM. (Detailed set-up in Topic 09.)

- 1 Confirm the required number of plots for the zone's area from the BAM plot table.
- 2 Place each plot in a representative, homogeneous part of one zone (never across two PCTs or condition states).
- 3 Record composition and cover in the 20×20 m quadrat and structural measures along the 50 m transect.
- 4 Log full plot metadata (ID, GPS, datum, bearing, date, observers, photos).

Worked example. A 3.2 ha zone of one PCT in moderate condition requires (per the BAM table) 2 plots. You place them in two representative patches away from edges and tracks, run both 20×20 m quadrats + 50 m transects, and capture metadata and photos for each.

METHOD 4 — TRANSECT (HOW TO)

Use it for: sampling structure or change along an environmental gradient (e.g. slope to drainage line).

- 1 Orient the transect along the gradient of interest; record start/end GPS and bearing.
- 2 Sample at set intervals or record intercepts (e.g. line-intercept for cover / coarse woody debris).
- 3 Note where structure or composition changes along the line to inform zone boundaries.

Worked example. A 50 m line-intercept from mid-slope woodland to a drainage line records where the shrub layer thickens and the ground flora shifts — confirming two vegetation zones rather than one, and feeding the coarse-woody-debris measure.

CHOOSING & COMBINING METHODS — WORKED BRIEF

Brief: 6 ha box-gum woodland lot for subdivision; desktop lists 3 candidate PCTs and 4 threatened plants (spring-flowering).

Method design: (1) *random meander* across all zones for the inventory; (2) *targeted searches* of mapped habitat for the 4 species, timed to their spring windows; (3) *BAM plots* — plot numbers per zone from the BAM table — for condition; (4) a *transect* where a gradient blurs a zone boundary.

Why adequate: every candidate PCT is plotted, every target species is searched in season with mapped effort, and all effort is logged — so the method statement will satisfy the BAM and the survey guidelines.

HOW TO RESEARCH & FIND THE INFORMATION

- The BAM plot table for required plot numbers per zone area.
- NSW/Commonwealth threatened-species survey guidelines for method, effort and timing.
- Target species' flowering/detectability windows in BioNet and our Threatened Flora ID guide.
- Previous reports for method precedents on similar sites.

DELIVERING THE OUTCOME — WHAT SUCCESS LOOKS LIKE

The deliverable is data collected by a **defensible, documented method**. Success means:

- A clear **methods statement** — what you did, how much, when, and why it was adequate — that the consent authority accepts **without an RFI**.
- Every zone plotted per the BAM and every target species searched in season, with **effort logged** (tracklogs, times, mapped search areas).
- Method chosen to **find the truth about the site**, not to minimise effort — our independent standard.

COMMON PITFALLS

- Under-surveying area or replication (too few plots/searches).
- Surveying at the wrong time of year for the target species.
- Meandering only the easy ground and missing microhabitats.
- Failing to record effort, so adequacy cannot be demonstrated.

CHECK YOUR UNDERSTANDING

- When would you use a meander vs a targeted search vs a plot?
- For the worked brief, how many method types would you run and why?
- What three things make a survey adequate, and how do you evidence each?

Related reading: Topic 02 (Objectives) sets what the method must achieve; [BAM Plot Guide](#) and Topic 09 detail plot set-up; the [Threatened Flora ID guide](#) holds the flowering calendar for timing.